Universal Design Project Portfolio



Min Wai
Electrical Engineering | University of Victoria | Email: minwai@uvic.ca |

Phone: 604-977-6091 | LinkedIn: linkedin.com/in/min-wai-a27442424

Universal Design Project - Enhanced Lighting and Security Alarm System

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About the Project

This portfolio summarizes my Design project, the Enhanced Lighting and Security Alarm System, developed for the University of Victoria Facilities Management design project. It is based on my final report and project planning documents.

Project Overview

The system combines security monitoring and smart lighting using a Raspberry Pi Pico W. It detects unauthorized entry using reed switches and a PIR motion sensor while allowing authorized access through an RFID scanner. LEDs provide lighting and visual status, and a buzzer alerts users during intrusions. The system was programmed in MicroPython and connected to a simple web interface.

Project Duration: January 2026 – April 2026

My Role

- Primary embedded software developer using Micro Python developed the main control logic.
- Designed and implemented the main control logic. Programmed LED and buzzer behaviour.
- Integrated RFID, PIR motion sensor, reed switch, LEDs, and buzzer.
- Supported webpage integration and real-time system status updates.
- Collaborated with teammates during system integration, testing, and debugging.
- Prepared technical documentation and contributed to design decisions throughout the engineering development process

Technologies

Hardware:

- Raspberry Pi Pico W
- RFID RC522
- PIR Motion Sensor
- Reed Switch
- LEDs
- Buzzer
- Breadboard
- GPIO

Software:

- MicroPython
- Thonny IDE

Key Features

- Detects unauthorized door opening.
- RFID bypass for authorized users.
- Motion-activated lighting after authorized entry.
- Visual and audible alarms during intrusion.
- Real-time monitoring through a web interface.

Key functions

Reed switch detects the door status during closing hours, while a PIR motion sensor monitors movement inside the building. LEDs provide both lighting and visual alarm indicators, and a buzzer rings when unauthorized entry is detected by the reed switch. The RFID scanner decides whether there is an authorized entry or not and controls the bypass mode, allowing the authorized individuals to temporarily disable the alarm.

All sensors and output devices are connected to the GPIO pins of the Pico board, where they are programmed in Micro Python and monitored through the user interface.

When a bypass is enabled, the system focuses on the motion-activated lighting function instead of triggering the alarm. If the PIR sensor detects any motion within every 30s interval after an authorized entry, the LEDs will stay on and if the motion is no longer detected, the LEDs will turn off.

If the door is opened without scanning the key card, the system will take it as an unauthorized entry and turn the LEDs on, trigger the alarm, and alert the user from the user interface.

Engineering Process

I followed an iterative engineering design process by researching solution options, developing prototypes, documenting results, testing each subsystem, and progressively integrating the complete system. Continuous testing, validation, and refinement improved system reliability while strengthening my engineering problem-solving and teamwork skills.

Challenges

The RFID module required additional libraries and careful hardware integration. Testing each sensor independently before combining them helped identify and resolve issues efficiently.

Testing & Validation

- Tested each sensor independently before system integration.
- Verified RFID authorization and alarm logic under different operating conditions.
- Validated motion-activated lighting functionality.

- Debugged hardware and software integration issues.
- Confirmed system operation through repeated functional testing.

Skills Demonstrated

- Engineering Design
- Embedded Systems
- MicroPython Programming
- Circuit Testing
- Technical Documentation
- Hardware & Software Integration
- Debugging & Troubleshooting
- Team Collaboration
- Problem Solving

Project Impact

The design was recommended as the team's final solution because of its strong accessibility, feasibility, low cost, and ability to integrate multiple security functions into one system.

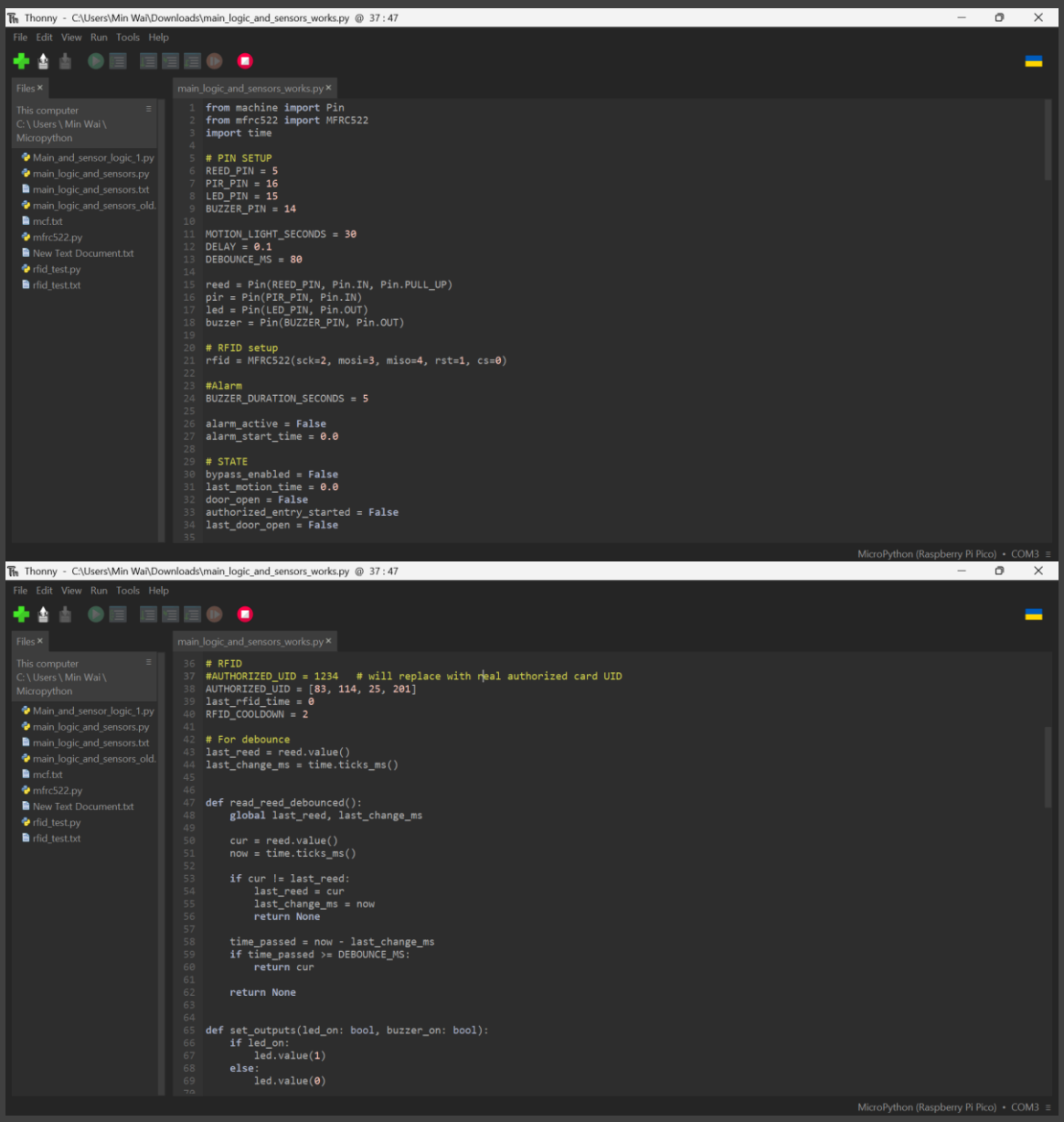
Feasibility

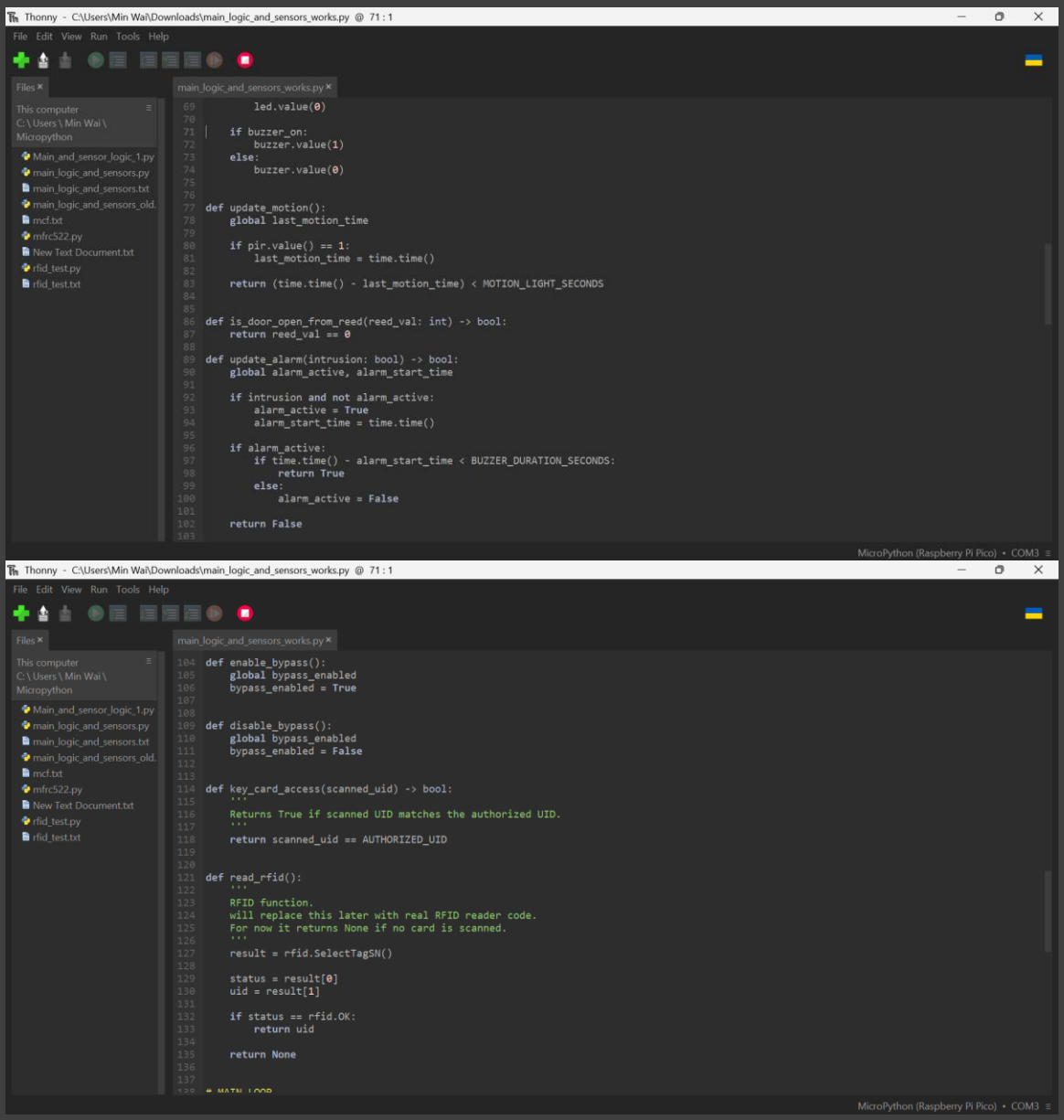
- Uses low-cost and widely available components such as Raspberry Pico, reed switch, PIR motion sensor, LEDs, buzzer, and RFID modules.
- Integrate sensors and output devices easily through GPIO pins on the Pico board using Micro Python.
- Requires simple hardware setup and minimal installation in a building.
- Processes sensor inputs using straightforward digital signals.
- Provides a practical solution for small residential or commercial environments.

How this Design meets the principles of universal design

- Clear alerts: LEDs and a buzzer provide both visual and audible warnings.
- Simple operation: Automatic sensors detect door opening and motion without user input.
- User flexibility: RFID bypass allows authorized users to disable the alarm when needed.
- Improved visibility: Motion-activated lighting increases safety during nighttime.
- Easy to understand: The system uses a simple interface and clear status indicator.

Project Photographs




Thonny - C:\Users\Min Wai\Downloads\main_logic_and_sensors_works.py @ 71 : 1
File Edit View Run Tools Help
Files
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Micropython
Main_and_sensor_logic_1.py
main_logic_and_sensors.py
main_logic_and_sensors.txt
main_logic_and_sensors_old.
mcf.txt
mfrc522.py
New Text Document.txt
rfid_test.py
rfid_test.txt
main_logic_and_sensors_works.py
led.value(0)
if buzzer_on:
buzzer.value(1)
else:
buzzer.value(0)
def update_motion():
global last_motion_time
if pir.value() == 1:
last_motion_time = time.time()
return (time.time() - last_motion_time) < MOTION_LIGHT_SECONDS
def is_door_open_from_reed(reed_val: int) -> bool:
return reed_val == 0
def update_alarm(intrusion: bool) -> bool:
global alarm_active, alarm_start_time
if intrusion and not alarm_active:
alarm_active = True
alarm_start_time = time.time()
if alarm_active:
if time.time() - alarm_start_time < BUZZER_DURATION_SECONDS:
return True
else:
alarm_active = False
return False
MicroPython (Raspberry Pi Pico) • COM3

Thonny - C:\Users\Min Wai\Downloads\main_logic_and_sensors_works.py @ 71 : 1
File Edit View Run Tools Help
Files
This computer
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Micropython
Main_and_sensor_logic_1.py
main_logic_and_sensors.py
main_logic_and_sensors.txt
main_logic_and_sensors_old.
mcf.txt
mfrc522.py
New Text Document.txt
rfid_test.py
rfid_test.txt
main_logic_and_sensors_works.py
def enable_bypass():
global bypass_enabled
bypass_enabled = True
def disable_bypass():
global bypass_enabled
bypass_enabled = False
def key_card_access(scanned_uid) -> bool:
'''
Returns True if scanned UID matches the authorized UID.
'''
return scanned_uid == AUTHORIZED_UID
def read_rfid():
'''
RFID function.
will replace this later with real RFID reader code.
For now it returns None if no card is scanned.
'''
result = rfid.SelectTagSN()
status = result[0]
uid = result[1]
if status == rfid.OK:
return uid
return None
MicroPython (Raspberry Pi Pico) • COM3

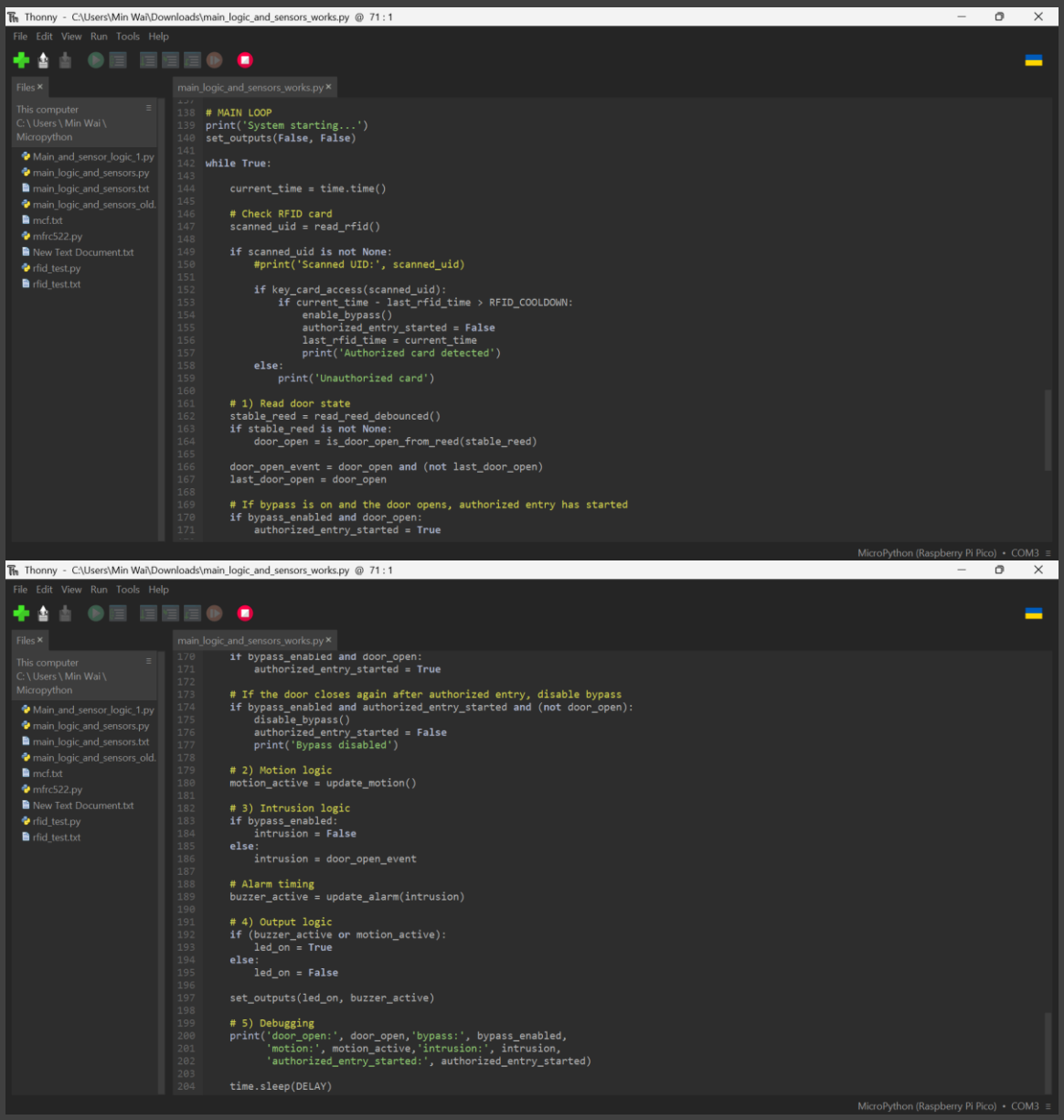


Figure 1: Main Control Logic Implementation (MicroPython)

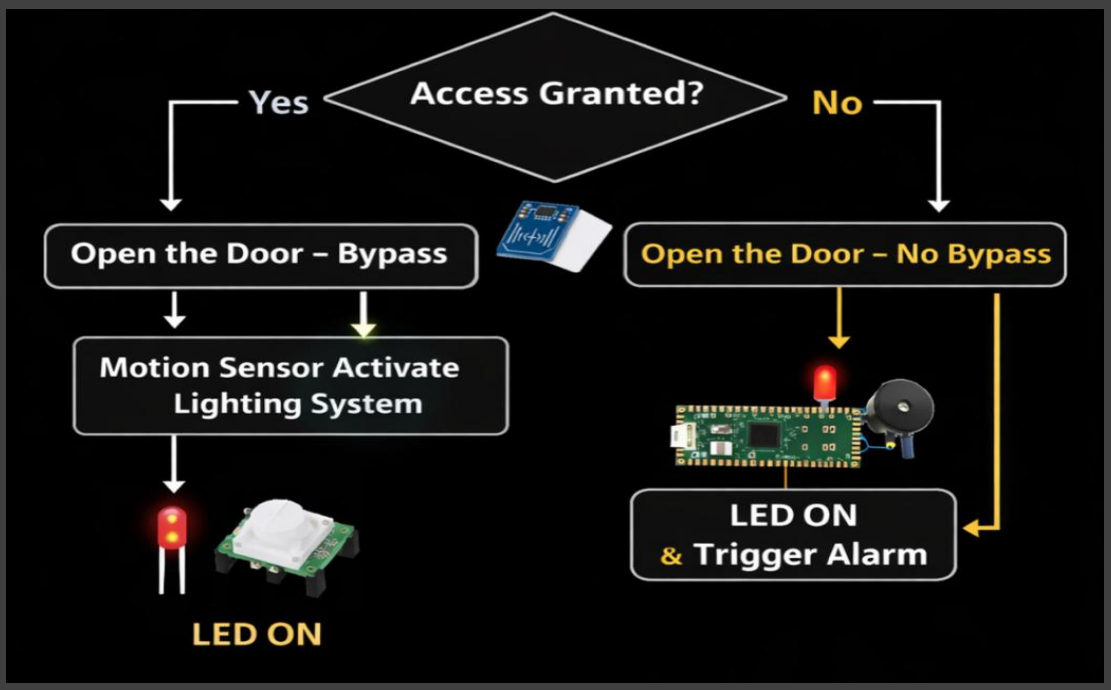


Figure 1.1: Main Control Logic Flowchart

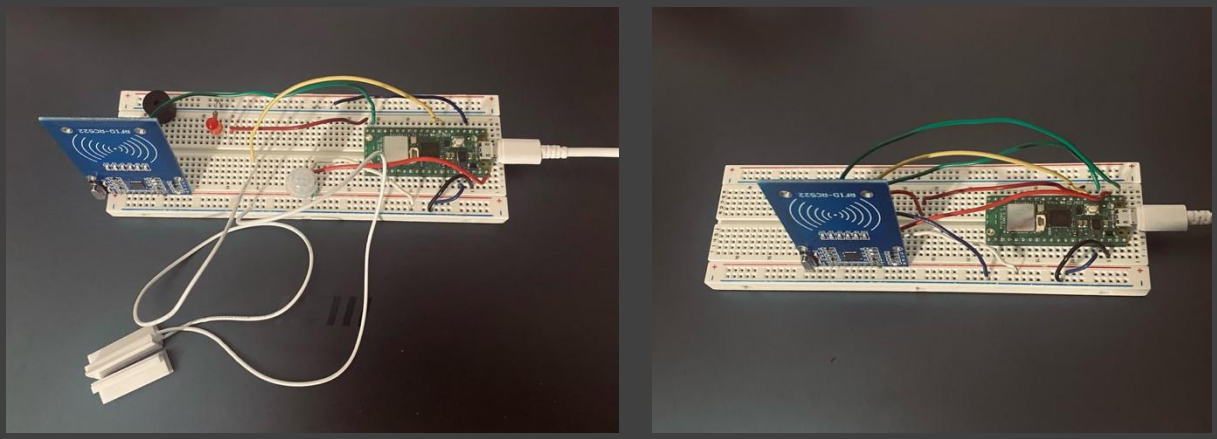


Figure 2.1: Physical Circuit Design

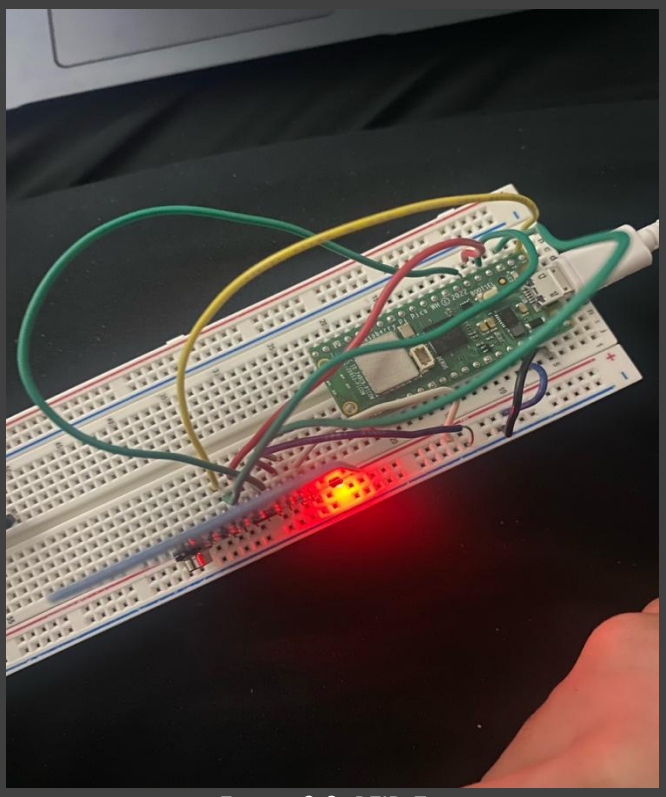


Figure 2.2: RFID Testing

What I learned

- Strengthened my engineering design and technical problem-solving skills through the development, testing, and refinement of an embedded system.
- Improved hardware and software integration through iterative testing.
- Gained practical experience troubleshooting embedded systems.
- Learned the importance of prototyping, validation, and teamwork during engineering design.
- Developed confidence presenting technical work and documenting engineering decisions.

References

Reference 1

Amir Mehdirezaei Khamse
Teaching Assistant
Department of Electrical & Computer Engineering
University of Victoria
Email: amehdirezaei@uvic.ca

Relationship: Teaching Assistant (ECE Lab – Section B09)

Reference 2

Ilamparithi Thirumarai Chelvan
Design Instructor
Department of Electrical & Computer Engineering
University of Victoria
Email: ilampari@uvic.ca

Relationship: Design Instructor (ENGR 120)